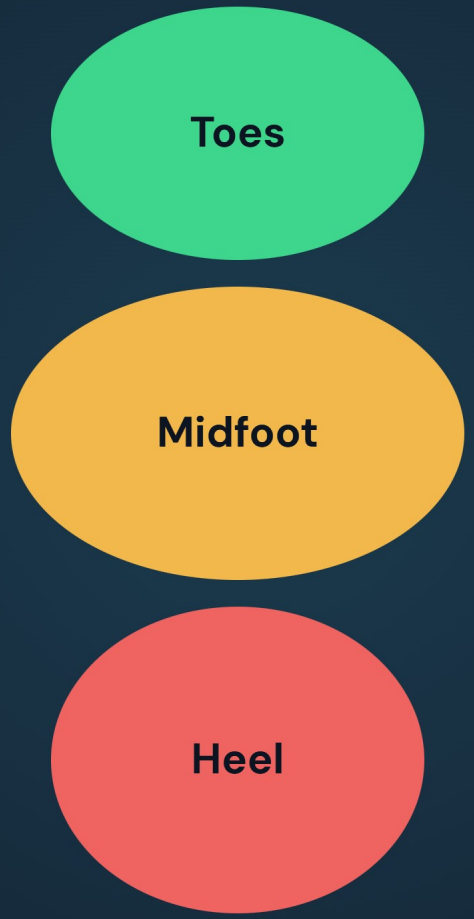
DIABETIC FOOT ULCER EARLY WARNING

UlcerSense

Explainable Digital Twin for Proactive Diabetic Foot Ulcer (DFU) Detection

Python 3.11 · OpenCV · XGBoost · SHAP · FastAPI



THE PROBLEM

Diabetic foot care starts after the skin has already broken.

Diabetic Foot Ulcers (DFUs) are among the leading causes of chronic infection, tissue necrosis and lower-limb amputation worldwide. Practice is mostly reactive: by the time ulceration or infection is visible, treatment is complex and costly, with high recurrence and amputation rates.

HEALTHCARE USE CASE

Hotspots appear before the wound does

TYPICAL HOTSPOT RISE

2.2–3.0 °C

above baseline or the contralateral foot site, driven by subcutaneous inflammation and impaired micro-vascular regulation.

TODAY · REACTIVE

Diagnosis follows skin breakdown, ulceration or infection, when treatment is complex, costly and prone to recurrence.

ULCERSENSE · PROACTIVE

A continuous, non-invasive digital twin flags thermal anomalies days before visible breakdown.

TECHNICAL STACK

Python 3.11 from thermogram to dashboard

COMPUTER VISION

OpenCV + NumPy

Isolates the plantar contour and maps pixels (0–255) to 26–36 °C.

FEATURE FUSION

Pandas

Merges thermal features with Age, HbA1c and diabetes duration.

PREDICTIVE ML

XGBoost

XGBClassifier outputs a calibrated ulceration risk, 0–100%.

EXPLAINABLE AI

SHAP

TreeExplainer names the primary risk driver for every inference.

BACKEND AND API

FastAPI + Uvicorn

Sub-second REST and streaming telemetry endpoints.

DIGITAL TWIN UI

Tailwind + Chart.js

HTML5 dark-mode dashboard with heatmap, risk curve and audit log.

AI/ML ENGINE

XGBoost predicts, SHAP explains

INPUT FEATURES

Thermal

Toes, Midfoot and Heel temperature

EHR

Age, HbA1c and diabetes duration

XGBCLASSIFIER

Gradient-boosted decision trees

Trained on localized spatial temperatures and systemic biomarkers.

OUTPUT AND EXPLANATION

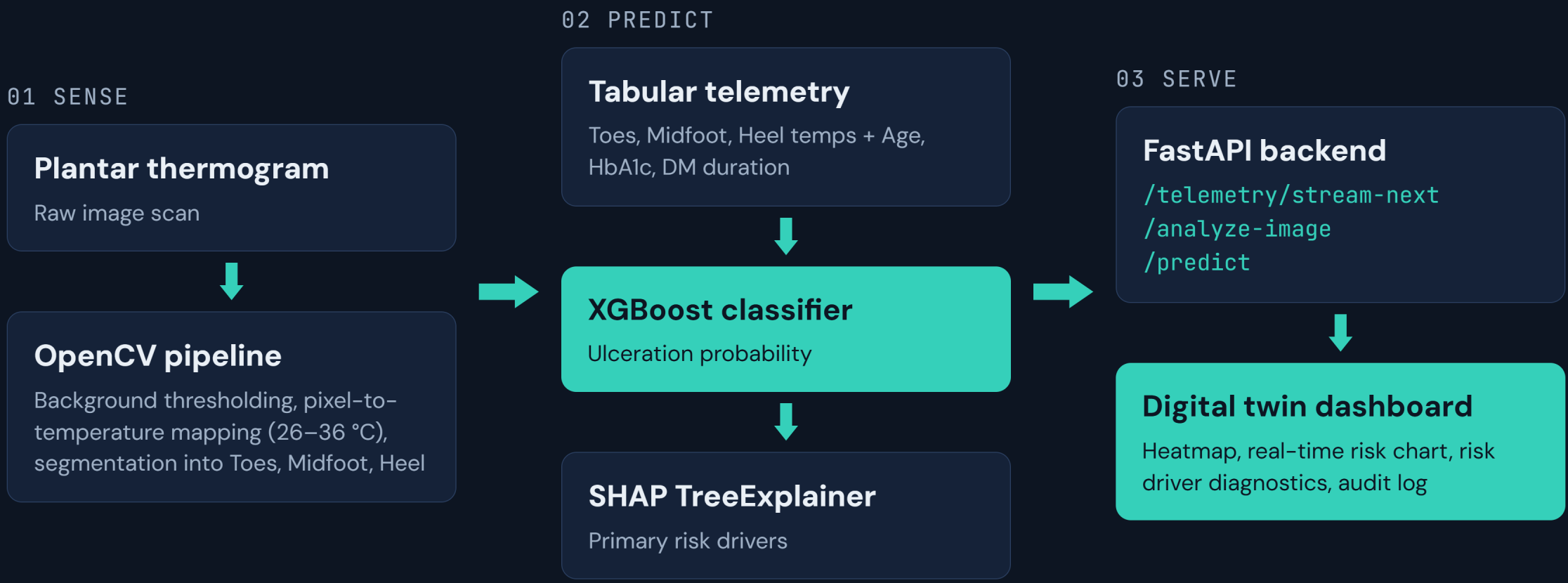
0%–100% risk

Calibrated probability, plus SHAP TreeExplainer attributions for every inference.

> Physicians see which biomarker drives the alert, such as heel temperature or elevated HbA1c, not just a score.

ARCHITECTURE

From thermogram to digital twin



FRONTEND INTERFACE

Digital twin dashboard

3-zone anatomical heatmap

Toes, Midfoot and Heel nodes shift from Green to Amber to Red.

Real-time risk line chart

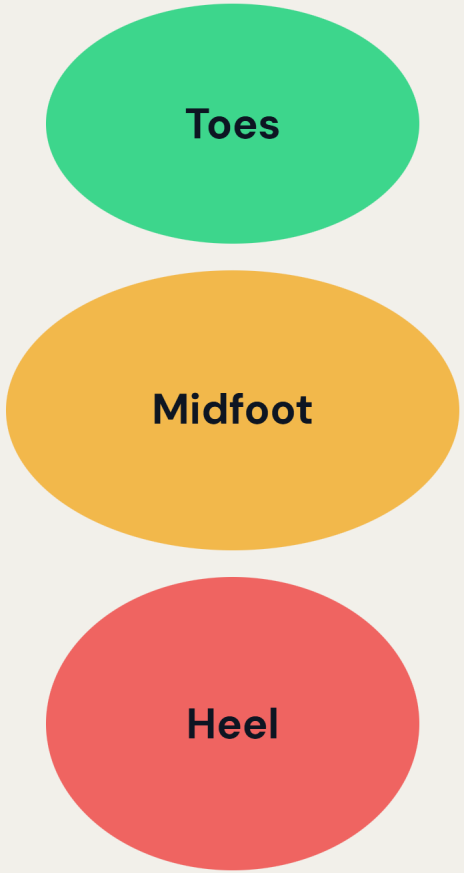
Continuous trajectory curves fed by the telemetry stream.

Risk contributor diagnostics

SHAP breakdown names the biomarker driving the risk.

History log and early-warning alerts

Audit trail of readings, with alerts when risk thresholds are crossed.



UNIQUE SELLING POINTS

What sets UlcerSense apart

Proactive, not reactive

Flags thermal anomalies days before visible breakdown.

Explainable by design

SHAP names the biomarker behind every risk score.

Thermal plus metabolic

Fuses regional foot temperature with Age, HbA1c and diabetes duration.

Live, non-invasive twin

Sub-second API and a clinical dashboard with early-warning alerts.

Moving diabetic foot management from reactive intervention to proactive, preventive wound care.